

## Graphical Abstract

### **Building Smarter IoT Systems: A Neuro-Symbolic Approach for Data Federation and Real-Time Processing**

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# Highlights

## Building Smarter IoT Systems: A Neuro-Symbolic Approach for Data Federation and Real-Time Processing

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- Formalized neuro-symbolic pipeline bridging deep perception with symbolic reasoning.
- Real-time federated integration via dynamic edge-to-cloud orchestration.
- High recognition accuracy (F1 89.4%) validated through 5-fold cross-validation.
- Scalable semantic layer with sub-50 ms query latency and reliable  $P_{95}$  performance.
- Self-correcting feedback loop reducing false positives to 3.9% in 4–5 cycles.
- Adaptive mechanism for mitigating neural hallucinations using grounded world models.

# Building Smarter IoT Systems: A Neuro-Symbolic Approach for Data Federation and Real-Time Processing

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## Abstract

The Internet of Things (IoT) generates vast volumes of heterogeneous and multimodal data, posing persistent challenges in semantic integration and real-time reasoning. Crucially, the effective synergy between these two pillars is what establishes a verifiable foundation for explainability in automated decision-making. Traditional AI approaches offer strong perceptual capabilities but lack transparency, while symbolic systems provide interpretability at the cost of real-time performance. This paper proposes a neuro-symbolic IoT pipeline that bridges the gap between deep neural perception and ontology-based symbolic reasoning. The architecture enables federated knowledge integration and a self-correcting feedback loop, delivering interpretable decisions across distributed IoT streams.

We validate the pipeline on two benchmark smart home datasets—CASAS and SPHERE—demonstrating strong activity recognition (F1 score up to 89.4%), low semantic query latency (<50 ms), fast complex event processing (CEP latency  $\leq 42$  ms), and a significant reduction in false positives through iterative feedback (down to 3.9%). Comparative studies show consistent gains in interpretability, correctness, and responsiveness over AI-only and KG-only baselines. These findings position our framework as an efficient, transparent, and adaptive foundation for real-time ambient intelligence in next-generation smart IoT systems.

**Keywords:** Internet of Things (IoT), Neuro-symbolic AI, Knowledge Graphs, Semantic Reasoning, Real-Time Inference, Semantic Integration

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## 1. Introduction

The rapid proliferation of the Internet of Things (IoT) has reshaped modern computing environments by interconnecting billions of heterogeneous devices capable of generating continuous, high-velocity data streams. These devices enable pervasive monitoring and intelligent automation across domains such as healthcare, autonomous transportation, manufacturing, and smart cities. However, the heterogeneous nature of IoT data—characterized by diverse formats (e.g., JSON, XML, binary payloads), inconsistent semantics, and varying levels of structure (ranging from rigidly structured telemetry to unstructured video streams)—poses significant challenges to large-scale data federation, semantic interoperability, and real-time processing [1]. The rapid expansion of IoT ecosystems further underscores the need to address these challenges in order to enable the deployment of reliable and efficient intelligent systems. From a system perspective, modern IoT applications demand not only accurate insights but also real-time responsiveness. Existing IoT data management approaches fall primarily under two distinct paradigms. On one hand, Knowledge-graph (KG)-based solutions provide structured, semantically rich representations that support unified data integration and machine-understandable interoperability [2, 3]. On the other hand, artificial intelligence (AI)-driven techniques, particularly

machine learning (ML) and deep learning (DL), contribute strong perceptual capabilities for pattern extraction in complex environments [4]. While KG-based methods enable explainable reasoning and semantic consistency, they struggle to process unstructured sensor data and maintain responsiveness in high-velocity scenarios. Conversely, AI-based systems demonstrate impressive scalability and adaptability but often operate as opaque “black-box” models with limited interpretability—an important gap in safety-critical IoT applications such as medical diagnosis and industrial automation. This dichotomy has motivated increasing interest in neuro-symbolic AI [5], a hybrid paradigm that integrates neural learning with symbolic reasoning to unify perception, semantics, and decision-making [6, 7]. Neuro-symbolic models enable high-level logical reasoning while leveraging deep neural networks to process unstructured and multimodal IoT data. This synergy provides both interpretability and adaptability, making it a compelling approach to address the limitations of standalone KG-based and AI-based solutions. From a system perspective, individual IoT applications demand not only accurate insights but also real-time responsiveness. Applications such as autonomous vehicles, remote patient monitoring, and industrial robotics rely on low-latency data processing and resilient decision pipelines. Traditional architectures often struggle to maintain performance as data volumes grow. Ensuring consistent processing capa-

bilities at scale—within a global landscape projected to exceed 39 billion connections by 2030 and surpass 50 billion by 2035 [8]—requires individual systems that seamlessly combine robust data federation with rapid neural inference.

### 1.1. Problem Statement

The central challenge addressed in this work is to reconcile the heterogeneity, dynamicity, and multimodality of IoT data within a unified processing architecture that combines neural perception, semantic integration, and symbolic reasoning—while ensuring transparent, low-latency decision-making and adaptive behavior in real-time environments. Existing knowledge-graph-based systems excel at semantic unification but lack mechanisms to handle unstructured sensor streams or adapt to dynamic environments. Meanwhile, AI-driven approaches efficiently process unstructured and high dimensional data but lack semantic grounding and explainability, which undermines trust and transparency in decision-making.

While recent efforts in other AI domains—such as Retrieval-Augmented Generation (RAG) successfully utilized multi-source neuro-symbolic integration to enhance reasoning and mitigate hallucinations, their application as cohesive architectures in real-time IoT environments remains limited. In particular, unifying neural perception, semantic interoperability, symbolic reasoning, and adaptive feedback under the constraints of continuous data streams and low-latency requirements is still insufficiently addressed. Bridging this gap is essential to enable interpretable, efficient, and responsive data federation in dynamic IoT environments.

### 1.2. Research Objectives

This study explores how a neuro-symbolic AI framework can address these limitations by integrating the strengths of neural and symbolic paradigms. The main research objectives are:

1. Introduce a neuro-symbolic pipeline that integrates neural models for unstructured IoT data processing with symbolic reasoning modules for semantic interoperability and explainable decision-making.
2. Design a real-time edge-to-cloud processing architecture that combines neural perception with knowledge-graph-based reasoning and CEP-style symbolic logic to improve low-latency decision-making while preserving transparency.
3. Evaluate federation efficiency, semantic reasoning correctness, and adaptive feedback performance, assessing the extent to which neuro-symbolic components improve throughput, accuracy, and symbolic consistency under real-world conditions.

The rest of this paper is structured as follows. Section 2 reviews related work across KG-based, AI-based, and neuro-symbolic IoT systems. Section 3 presents the proposed neuro-symbolic pipeline and its architectural modules. Section 4 details the experimental design, datasets, and evaluation metrics.

Section 5 discusses results, insights, and limitations. Section 6 concludes with a summary of contributions and future directions.

## 2. Related Work

This section reviews the existing literature on data federation and real-time data processing in IoT environments. It highlights key contributions and identifies limitations that motivate the proposed architecture.

### 2.1. IoT Data Federation

IoT data federation aims to integrate heterogeneous and distributed data sources into a unified representation that supports efficient querying, analytics, and semantic interpretation. Traditional federation models range from centralized architectures to distributed edge-to-cloud ecosystems, yet they often fail to accommodate the increasing variety, volume, and velocity of IoT data. To address these limitations, two major categories have emerged: knowledge-graph-based approaches and hybrid AI/neuro-symbolic approaches.

*KG-Based Approaches.* Knowledge graphs (KGs) provide a structured semantic layer capable of modeling heterogeneous IoT data and unifying device-level information. Wickramarachchi et al. [9] demonstrated the effectiveness of KG embeddings for autonomous driving data, enabling improved relational modeling of sensor streams. Van der Weerd et al. [10] applied KG-based transformation pipelines for smart homes, where IoT data was converted into RDF and integrated into predictive models to improve forecaster performance. In industrial contexts, Yang et al. [11] introduced the InPro workflow ontology to track and synchronize material flows across production environments, while Li et al. [12] constructed a medical KG from real clinical records to support diagnostic reasoning. In addition, Diamantini et al. [13] proposed the Process-aware Industrial Internet of Things (IIoT) Knowledge Graph, a semantic integration framework that enriches industrial sensor streams with process-level ontological descriptions to support context-aware analytics. While highly effective for structured IIoT environments, the approach does not incorporate neural perception or handle unstructured sensor modalities.

These studies highlight the value of semantic modeling for IoT interoperability. However, KG-based systems exhibit well-known limitations:

- Difficulty in processing unstructured sensors (video, audio, time-series),
- High computational overhead in maintaining large dynamic KGs,
- Limited efficiency under real-time ingestion constraints.

Thus, while KGs provide strong semantic grounding, they require complementary approaches to handle perception and real-time processing efficiently.

*AI and Neuro-Symbolic Approaches.* AI-based methods—especially deep learning—offer strong capabilities for processing high-dimensional and noisy IoT data. In the healthcare domain, Zhao et al. [14] used neuro-symbolic pipelines to build medical KGs from Electronic Medical Records (EMRs) using BiLSTM-CRF models, while Rivas et al. [15] demonstrated how hybrid neuro-symbolic reasoning enhances KG-based link prediction and improves transparency in clinical settings. Beyond healthcare, Gilpin and Ilievski [16] applied neuro-symbolic techniques to traffic management, combining neural prediction modules with symbolic logic constraints. More recently, Amlashi et al. [17] proposed a neuro-symbolic architecture for automated IoT platform configuration, combining conceptual modeling with symbolic rule specification and neural components for device discovery. Their work shows the feasibility of neuro-symbolic reasoning in complex IoT environments, but it focuses on configuration management rather than heterogeneous data federation, dynamic semantic reasoning, or feedback adaptation in real-time contexts.

Neuro-symbolic approaches address key limitations in pure AI systems by introducing explainability, enhancing generalization through structured knowledge, and improving data integration via semantic grounding. However, existing works are often tailored to specific domains and lack general-purpose frameworks applicable to real-time IoT federation or feedback-driven adaptation.

## 2.2. Real-Time Data Processing in IoT

Real-time analytics is essential for latency-sensitive IoT applications such as connected vehicles, industrial automation, and medical monitoring.

*KG-Based Real-Time Processing.* KGs have been applied in real-time scenarios primarily for structured semantic reasoning. Zahra et al. [18] proposed a semantic modeling approach for Industry 4.0 capable of monitoring environmental conditions. Zhao et al. [19] introduced a dynamic KG for digital-twin-enabled production logistics, ensuring spatial-temporal consistency in decision-making. While effective for structured reasoning, these methods struggle with high-velocity updates, efficiency of reasoning operations, and integration of unstructured data streams. More recently, Khan et al. [20] introduced KEM-IoMT, a knowledge-graph-embedding-enhanced framework for medical IoT recommendation tasks. The model constructs a KG from heterogeneous medical and contextual data and uses GNN-based embeddings to support accurate service recommendations. Although the work demonstrates the potential of KG-driven analytics in IoT-derived domains, it does not address multimodal data fusion, real-time reasoning constraints, or feedback-based learning.

*AI and Neuro-Symbolic Real-Time Processing.* Deep learning models have demonstrated strong performance in real-time IoT analytics. Hossam et al. [21] proposed a DL-based smart city framework addressing heterogeneity and decision latency.

Bulla et al. [22] incorporated deep models for root-cause analysis in fog environments but faced compute scalability issues.

To overcome interpretability limitations, neuro-symbolic methods have emerged. Sharifi et al. [23] integrated symbolic logic with deep reinforcement learning for autonomous driving, improving both safety constraints and real-time performance. Yu et al. [24] developed a neuro-symbolic healthcare system for chronic disease management, showing the feasibility of combining KG reasoning with neural models in time-critical applications. Expanding into real-time constraints and resource optimization, recent works have demonstrated the viability of neuro-symbolic AI at the edge. Badhan [25] introduced a lightweight neuro-symbolic framework for interpretable IoT edge intrusion detection, utilizing knowledge distillation to reduce inference latency. Similarly, Goswami et al. [26] proposed a neuro-symbolic reasoning approach combining graph neural networks with symbolic rules for robust stochastic power allocation in Industrial IoT federated learning. While these works confirm the viability of neuro-symbolic reasoning in complex, resource-constrained IoT environments, they focus strictly on network security and power optimization tasks, rather than addressing heterogeneous data federation, dynamic semantic reasoning, or multimodal feedback adaptation.

## 2.3. Synthesis and Research Gap

Based on the review above, Table 1 provides synthesis of the current research landscape. It summarizes the limitations of existing works to clearly highlight the scientific gap that our neuro-symbolic pipeline aims to address:

- KG-based systems offer strong interoperability but struggle with real-time processing.
- AI-based systems handle high-velocity data but lack interpretability and semantic consistency.
- Neuro-symbolic systems combine both strengths, yet current work does not fully support large-scale federation, multimodal integration, or adaptive reasoning.

While prior evaluations are often domain-specific or rely on synthetic workloads, our approach demonstrates broader applicability using two real-world benchmarks—CASAS and SPHERE—featuring heterogeneous sensors and annotation schemes. We introduce a unified neuro-symbolic IoT pipeline that integrates neural activity recognition, RDF-based semantic modeling, SWRL-based reasoning, and adaptive feedback, enabling interpretable, real-time reasoning across diverse smart environments.

## 3. Neuro-Symbolic IoT Pipeline

This section presents the proposed *Neuro-Symbolic IoT Pipeline*, a unified architecture that integrates neural perception, semantic data federation, symbolic reasoning, and real-time feedback mechanisms. The pipeline is structured into three interconnected stages—data acquisition and neural perception, semantic representation and federation, and neuro-symbolic reasoning and feedback—as illustrated in Fig. 1.

The objective of this design is to simultaneously address three central challenges in IoT intelligence: the need to process

Table 1: Synthesis of existing IoT data federation and real-time data processing approaches: Categorization and identified gaps.

| Paper                         | Approach       | Domain                              | Limitations                                                                  |
|-------------------------------|----------------|-------------------------------------|------------------------------------------------------------------------------|
| Wickramarachchi et al. (2020) | KG-based       | Autonomous Vehicles                 | Difficulty handling unstructured sensor data and real-time updates.          |
| Van der Weerd et al. (2021)   | KG-based       | Smart Homes                         | Scalability limitations when processing large-scale IoT streams.             |
| Yang et al. (2023)            | KG-based       | Industrial Automation               | Scalability issues in highly dynamic production environments.                |
| Li et al. (2020)              | KG-based       | Healthcare                          | Challenges integrating unstructured data and supporting real-time reasoning. |
| Diamantini et al. (2023)      | KG-based       | Industrial IoT                      | Lacks neural perception; scalability issues under real-time conditions.      |
| Khan et al. (2025)            | KG-based       | IoMT                                | Lacks multimodal fusion and real-time feedback adaptation.                   |
| Hossam et al. (2024)          | AI-based       | Smart Cities                        | Limited interpretability for high-stakes real-time decisions.                |
| Bulla et al. (2022)           | AI-based       | Industrial Automation               | Scalability and transparency challenges as system complexity increases.      |
| Zhao et al. (2021)            | Neuro-symbolic | Healthcare                          | Challenges in scalability and maintaining transparent reasoning.             |
| Gilpin & Ilievski (2021)      | Neuro-symbolic | Autonomous Vehicles                 | Difficulty explaining real-time logic; lacks adaptive learning.              |
| Amlashi et al. (2025)         | Neuro-symbolic | IoT Configuration                   | Focuses on static setup rather than dynamic data federation.                 |
| Rivas et al. (2022)           | Neuro-symbolic | Healthcare                          | Scalability issues; lacks an automated feedback correction mechanism.        |
| Sharifi et al. (2023)         | Neuro-symbolic | Autonomous Vehicles                 | Issues in handling high-velocity data while ensuring interpretability.       |
| Badhan (2026) [25]            | Neuro-symbolic | IoT Edge Security                   | Focuses on intrusion detection rather than multimodal reasoning.             |
| Goswami et al. (2026) [26]    | Neuro-symbolic | Industrial IoT (Federated Learning) | Restricted to power allocation; lacks semantic integration.                  |

heterogeneous multimodal data, the importance of maintaining an interpretable semantic layer, and the requirement for real-time reasoning capable of adapting to dynamic environments.

### 3.1. Formal Problem Definition

The Neuro-Symbolic IoT Pipeline addresses the general problem of transforming heterogeneous, multimodal IoT data streams into semantically grounded, interpretable and real-time decisions. To formalize this process, let  $X_t = \{x_1, x_2, \dots, x_n\}$  denotes the collection of raw observations generated at time  $t$  by a diverse set of IoT devices. Each observation may be a numerical reading, an image frame, a point-cloud segment, or a structured log record. A neural perception function  $f_\theta(\cdot)$  parameterized by weights  $\theta$  maps each observation to an embedding  $z_t = f_\theta(x_t)$  that captures features relevant to downstream reasoning. In parallel, the semantic layer maintains an evolving knowledge graph  $G_t = (V_t, E_t)$  whose nodes represent entities or concepts and whose edges represent relations or events inferred from the IoT environment. Symbolic rules and ontological constraints, denoted by the set  $R$ , encode domain knowledge and regulate permissible inferences. The reasoning objective is to infer a set of interpretations  $Y_t$  such that  $(G_t, z_t, R) \mapsto Y_t$ , where  $Y_t$  may contain semantic classifications, detected events,

recommended actions, or explanatory statements. This formalization establishes a mathematical framework to systematically map the pipeline’s inputs and state transitions to the empirical evaluation metrics. Specifically, the quality of the inferred interpretations  $Y_t$  and the efficiency of applying rules  $R$  are quantitatively assessed in Section 4.4 using metrics such as classification accuracy, rule-based inference correctness, and feedback latency.

### 3.2. Architectural Overview

The pipeline begins with raw sensor information originating from diverse IoT devices, including cameras, LIDAR units, UAV-mounted platforms, and system logs generated in formats such as JSON and XML. These data streams are first processed by the **Data Collection and Preprocessing** module, which normalizes and harmonizes them into representations suitable for computation. During this step, raw inputs are first transformed into engineered feature vectors and temporal sequences (representing localized signal properties), which are subsequently formatted into unified preprocessed tensors (serve as the standard computational structure) for subsequent neural analysis. This architecture was instantiated and evaluated using real-world smart home datasets (CASAS and SPHERE), as described in

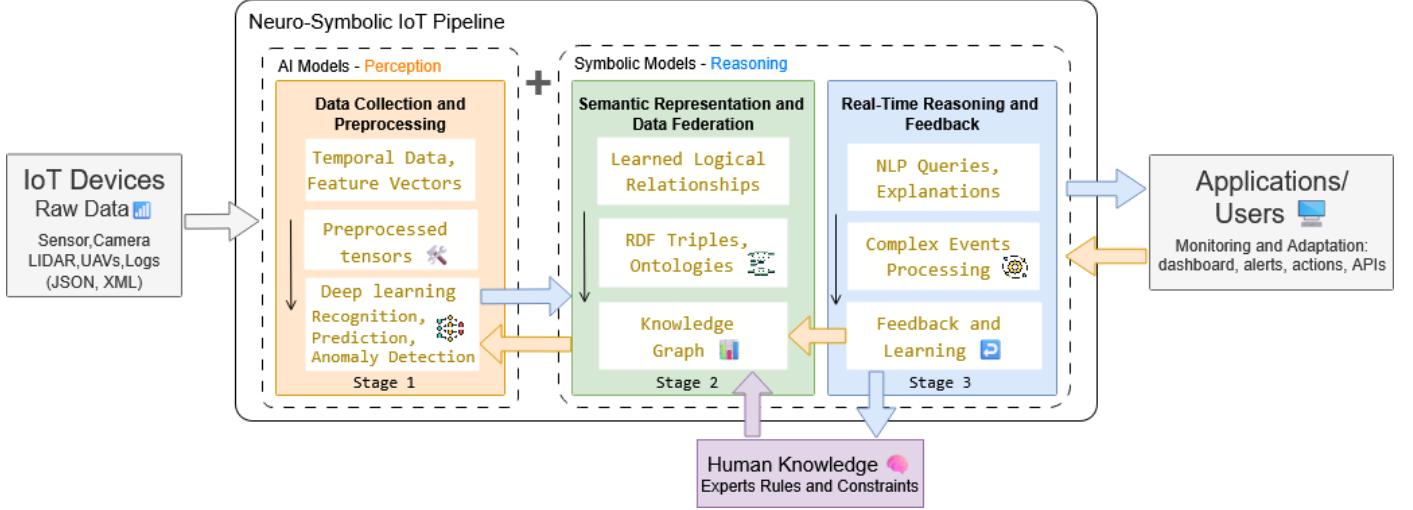


Figure 1: Overview of the proposed **Neuro-Symbolic IoT Pipeline**.

#### Section 4.1.

Once these representations are obtained, the **AI Models (Perception)** layer performs high-level interpretation of the sensor streams. Deep learning models extract salient patterns, whether spatial, temporal, or contextual, generating neural encodings that serve both as direct evidence for inference and as semantic signals that enrich the symbolic layer. These neural outputs are passed into the **Semantic Representation and Data Federation** module, where they are aligned with domain ontologies, converted into RDF triples, and incorporated into a unified **Knowledge Graph** that maintains an evolving view of the IoT ecosystem.

Building on this semantic foundation, the **Symbolic Models (Reasoning)** layer performs structured inference operations. It integrates learned logical relationships, domain constraints, and human knowledge to generate interpretable responses. Through techniques such as rule-based reasoning and Complex Event Processing (CEP), the system identifies situational patterns, resolves context, and produces outputs such as explanations, recommended actions, or responses to user queries. These outputs are then conveyed to **Applications/Users** via dashboards, alerts, or API-driven services.

The final component of the architecture is the **Real-Time Reasoning and Feedback** mechanism. This module closes the loop by monitoring system behavior and operational outcomes, updating neural and symbolic modules accordingly, and refining system performance over time. Through continuous feedback and learning, the pipeline maintains resilience to evolving conditions and supports low-latency, adaptive IoT decision-making—a claim we empirically validate in Section 4.4.4, demonstrating that stage-specific latencies remain strictly under 80ms despite the multi-step architecture.

##### 3.2.1. Stage 1: Neural Perception

The initial stage of the pipeline focuses on consolidating data produced across a wide spectrum of IoT devices. Cameras and video sources supply visual information, LIDAR and 3D

point-cloud sensors offer spatial awareness, UAV-based platforms capture aerial perspectives, and system logs contribute time-series telemetry. Because these data types vary significantly in structure and timing, the preprocessing stage plays a critical role in establishing a consistent computational foundation.

Within the **Data Collection and Preprocessing** module, raw signals are filtered, synchronized, and gradually transformed into structured inputs. Formally, this workflow is defined as a sequence of three mapping functions:

1. A raw heterogeneous signal  $x_i^{\text{raw}}$  is first filtered and time-aligned via a synchronization function  $\phi_{\text{sync}}$ , yielding an intermediate synchronized state  $x_i^{\text{sync}} = \phi_{\text{sync}}(x_i^{\text{raw}})$ .
2. A feature extraction mapping  $\phi_{\text{feat}}$  then identifies relevant patterns to produce engineered **feature vectors** and temporal sequences  $x_i^{\text{feat}} = \phi_{\text{feat}}(x_i^{\text{sync}})$ .
3. Finally, a formatting function  $\phi_{\text{format}}$  structures these vectors into high-dimensional **tensors**  $x_i^{\text{tensor}} = \phi_{\text{format}}(x_i^{\text{feat}})$ , ensuring they are computationally compatible with the deep learning models.

This gradual transformation ensures that downstream components operate on semantically coherent and structurally compatible inputs. By resolving format inconsistencies early in the pipeline, this stage ensures that downstream components operate on semantically coherent and structurally compatible inputs.

The resulting data are then processed by the **AI Models (Perception)**. Here, deep neural networks trained on tasks such as image recognition, spatiotemporal prediction, and anomaly detection transform input tensors into compact latent features, temporal embeddings, and descriptive state representations. These models capture correlations and patterns that are either too subtle (e.g., micro-variations in wearable accelerometer frequencies) or too complex (e.g., high-dimensional spatiotemporal dependencies across continuous video frames) to express

symbolically at the raw-data level. Once extracted, the neural features serve two complementary purposes: (1) they enrich the semantic representation layer and (2) provide evidential grounding for higher-level symbolic reasoning.

### 3.2.2. Stage 2: Semantic Federation

To achieve interoperability across heterogeneous sources, the pipeline incorporates a dedicated **Semantic Representation and Data Federation** module. This component converts neural outputs, temporal features, and relevant metadata into structured knowledge elements that populate a unified semantic view of the system. RDF triples are generated to describe device states, events, and relationships, while domain ontologies encode conceptual constraints and contextual meaning. These resources collectively form the **Knowledge Graph**, which acts as the centralized semantic backbone of the architecture.

The knowledge graph supports expressive querying, long-term contextual tracking, and structured reasoning. By associating neural features with symbolic concepts, the system bridges low-level perception with high-level understanding, enabling interpretability and semantically grounded decision-making. For example, if the neural module detects a rapid posture transition (e.g., *Standing*  $\rightarrow$  *Lying*) from raw accelerometer data, the knowledge graph contextualizes this event with spatial and temporal entities (e.g., `locatedIn(Person, Bathroom)` at 02:00 AM). The symbolic engine can then logically infer a *Fall Alert*, providing a fully interpretable trace of both the neural evidence and the ontological rules that led to the decision. To formalize this transformation from the continuous neural vector space to the discrete symbolic graph space, Algorithm 1 illustrates the semantic federation process. It details how raw metadata and neural outputs are converted into RDF assertions and validated against the domain ontology before updating the Knowledge Graph.

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#### Algorithm 1 Neural-to-Semantic Federation Process

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**Require:** Neural embeddings  $z_t$ , Predicted classes  $\hat{y}_t$ , Confidence scores  $p_t$ , Metadata  $m_t$  (Device ID, Timestamps), Ontology  $\mathcal{O}$ , Current KG  $G_t$

**Ensure:** Updated Knowledge Graph  $G_{t+1}$

```

1:  $\mathcal{T} \leftarrow \emptyset$  ▷ Initialize empty set of RDF triples
2: for each instance  $i$  in current batch do
3:    $e_{\text{dev}} \leftarrow \text{ResolveURI}(m_t[i].\text{deviceId}, \mathcal{O})$ 
4:    $e_{\text{time}} \leftarrow \text{FormatXSDTime}(m_t[i].\text{timestamp})$ 
5:    $e_{\text{act}} \leftarrow \text{ResolveURI}(\hat{y}_t[i], \mathcal{O})$ 
▷ Generate Structural and Temporal RDF Triples
6:    $\mathcal{T} \leftarrow \mathcal{T} \cup \{(e_{\text{dev}}, \text{hasPredictedState}, e_{\text{act}})\}$ 
7:    $\mathcal{T} \leftarrow \mathcal{T} \cup \{(e_{\text{act}}, \text{occursAt}, e_{\text{time}})\}$ 
▷ Attach Neural Context (Loose Coupling)
8:    $\mathcal{T} \leftarrow \mathcal{T} \cup \{(e_{\text{act}}, \text{hasLatentVector}, z_t[i])\}$ 
9:    $\mathcal{T} \leftarrow \mathcal{T} \cup \{(e_{\text{act}}, \text{hasConfidence}, p_t[i])\}$ 
10: end for
11: if CheckOntologicalConsistency( $\mathcal{T}, \mathcal{O}$ ) == True then
12:    $G_{t+1} \leftarrow G_t \cup \mathcal{T}$  ▷ Ingest into Knowledge Graph
13: else
14:   LogConflict( $\mathcal{T}$ ) ▷ Flag for Runtime Monitor /
Feedback
15:    $G_{t+1} \leftarrow G_t$ 
16: end if
17: return  $G_{t+1}$ 

```

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### 3.2.3. Stage 3: Reasoning and Feedback

The **Symbolic Models (Reasoning)** layer ensures inference tasks that require explicit logic. It assesses the semantic graph to identify relevant patterns, validates consistency using ontological axioms, and applies CEP to detect time-dependent sequences. Through this reasoning cycle, the system generates explanations, responds to natural language queries, and issues actionable recommendations. Human knowledge may also be incorporated into the reasoning process. This allows experts to define symbolic rules and ontological constraints that guide the reasoning process, ensuring that the system’s decisions are aligned with verifiable human logic rather than relying solely on black-box neural predictions.

A core innovation of the proposed pipeline lies in its ability to maintain responsiveness through continuous adaptation. The **Real-Time Reasoning and Feedback** module observes system outputs in parallel with environmental conditions, allowing it to refine symbolic rules, enrich the knowledge graph with new contextual insights, and update neural models using supervised or semi-supervised signals.

Feedback is triggered by semantic contradictions (e.g., inferred activity contradicts sensor inactivity), out-of-distribution neural predictions, or violations of symbolic preconditions (e.g., expected room occupancy not satisfied). These inconsistencies are detected by runtime monitors—implemented as continuous SPARQL constraint queries that aggregate and validate requirements across all federated sensor sources—observing both KG assertions and neural classifier outputs. Upon detection, the system selectively initiates actions such as symbolic rule refinement, retraction of incorrect RDF assertions, or tar-



geted retraining of the neural module on recent error cases. Furthermore, as purely automated adaptation carries risks, this stage supports a Human-in-the-Loop (HITL) mechanism; domain experts can intervene early to validate flagged contradictions, ensuring that rule refinements and neural retraining are both safe and semantically accurate before being fully integrated. To formalize this continuous adaptation process, Algorithm 2 details the execution cycle of the runtime monitors and the resulting feedback loop.

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**Algorithm 2** Real-Time Neuro-Symbolic Feedback Loop

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**Require:** Data stream  $X_t$ , Neural model  $f_\theta$ , Knowledge Graph  $G_t$ , Rule base  $R$

**Ensure:** Interpreted events  $Y_t$ , Updated model  $f_{\theta'}$ , Updated Graph  $G_{t+1}$

```

1: loop
2:    $x_t \leftarrow \text{ReadNextBatch}(X_t)$ 
3:    $z_t, p_t \leftarrow f_\theta(x_t)$   $\triangleright$  Extract latent features & predictions
4:    $G'_t \leftarrow \text{MapToRDF}(G_t, z_t, p_t)$   $\triangleright$  Semantic federation
5:    $Y_t \leftarrow \text{CEP\_Engine}(G'_t, R)$   $\triangleright$  Symbolic reasoning
6:   if  $\text{CheckContradiction}(Y_t, p_t, R) == \text{True}$  then
7:      $G_{t+1} \leftarrow \text{RetractFalseTriples}(G'_t)$ 
8:      $R \leftarrow \text{RefineRules}(R, \text{HITL\_Validation})$ 
9:      $f_{\theta'} \leftarrow \text{PartialRetrain}(f_\theta, x_t, Y_t^{\text{corrected}})$ 
10:  else
11:     $G_{t+1} \leftarrow G'_t$ 
12:     $f_{\theta'} \leftarrow f_\theta$ 
13:  end if
14:  Output  $Y_t$   $\triangleright$  Publish the interpreted event
15:   $f_\theta \leftarrow f_{\theta'}$   $\triangleright$  Apply model updates for next iteration
16:   $t \leftarrow t + 1$   $\triangleright$  Advance time step
17: end loop

```

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The architecture also incorporates **Monitoring and Adaptation** interfaces that expose insights to applications and users through dashboards, alerts, and API endpoints. These interfaces surface system behavior, collect runtime observations, and deliver explanations or recommended actions derived from symbolic reasoning. Feedback from these interactions informs subsequent adaptive updates, enhancing robustness to concept drift and preserving performance under fluctuating or uncertain conditions.

### 3.3. Data Flow and Module Interaction

Although the pipeline is composed of distinct modules, its effectiveness relies on the fluid interaction between neural and symbolic processes. As illustrated in Fig. 2, the data flow begins at the IoT devices, where raw sensor streams are transformed into normalized representations that ensure compatibility across modalities. These representations enter the perception layer, where neural models derive latent features that capture patterns and correlations not explicitly available in symbolic form. Rather than treating these neural outputs as isolated predictions, the pipeline integrates them directly into the semantic federation stage, enriching RDF triples, grounding on-

tological assertions, and updating the knowledge graph with context-aware descriptions of system states.

To preserve modularity and transparency, neural outputs are not hardcoded as RDF facts but linked via lightweight semantic annotations—such as predicted activity, temporal scope, and confidence level. This loose coupling allows the symbolic layer to reason over perceptual results while retaining traceability and the ability to revise or override neural inferences. Conversely, symbolic inconsistencies or rule violations can propagate corrective signals backward, triggering targeted neural retraining or refinement of latent features. This bidirectional interaction ensures continuous system evolution: neural models provide the evidential grounding for logical reasoning, while symbolic rules identify inconsistencies to trigger targeted neural retraining. By utilizing standardized communication interfaces (such as RDF triples and confidence scores), this feedback loop enables adaptive intelligence while keeping the perception and reasoning modules structurally independent and modular.

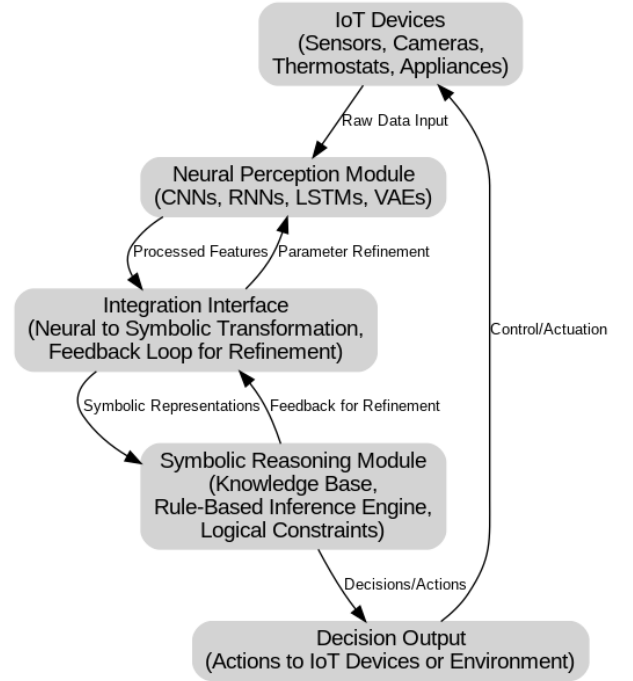


Figure 2: Neuro-symbolic data flow and module interaction.

Once the semantic layer has been updated, symbolic reasoning operates on a combination of structured KG representations and neural-derived evidence. This hybrid inference process enables the system to detect events whose meaning emerges from both quantitative patterns and semantic constraints. For example, a neural model may detect anomalous movement or unusual temporal patterns, while the symbolic module interprets these findings in the context of domain rules, safety thresholds, and spatial or temporal relationships encoded within the ontology. The interpretation then triggers actions, explanations, or alerts delivered to users or automated processes.

Feedback plays an equally important role in the interaction flow. Whenever the system detects inconsistencies, false

positives, or evolving environmental conditions, corrective insights are propagated back through the pipeline. Symbolic rules may be refined, KG assertions updated, and neural models partially retrained or recalibrated. This integration transforms the pipeline into a continuous, self-correcting cycle (unified cognitive loop) where perception and reasoning are mutually reinforcing. By documenting every decision through symbolic traces, the system provides intrinsic explainability, while the automated feedback loop ensures operational robustness by enabling the architecture to detect and resolve internal contradictions autonomously.

### 3.4. Illustrative Use Case: Temporal Reasoning and Anomaly Detection

To ground the theoretical pipeline in a practical reality, consider a smart home anomaly detection scenario. A critical dimension of this process is the handling of time, which is operationalized throughout the pipeline via timestamped data streams and sliding window analysis.

When raw sensors (e.g., motion and appliance monitors) emit signals, the preprocessing module aggregates them into fixed-length temporal windows (e.g., 30 seconds). The **Neural Perception** module processes these windows, predicting, for instance, that a "Meal Preparation" activity is occurring at  $t = 03:00$  AM. During **Semantic Federation**, this prediction is converted into time-annotated RDF triples, using temporal predicates such as `occursAt(Event, "03:00:00")`.

The **Symbolic Reasoning** engine utilizes CEP to maintain a short-term memory of these time-annotated triples across a sliding window (e.g., the last 5 minutes). It evaluates this temporal context against domain rules—such as an ontological constraint stating that cooking activities are highly anomalous during designated sleep hours. If the engine detects this temporal conflict, it immediately flags a night-time anomaly. Furthermore, if the system later confirms this was a false positive caused by sensor noise, the **Feedback** loop uses the exact temporal boundaries of the event to extract the faulty sensor segment, retract the erroneous RDF triples, and trigger a targeted neural retraining update.

### 3.5. Execution Model and Computational Considerations

Given the distributed nature of IoT environments, the execution of the pipeline is orchestrated across edge, fog, and cloud layers to balance latency, computational load, and scalability. Lightweight preprocessing and initial neural inference are executed at edge or fog nodes, where proximity to sensors reduces communication delays and ensures timely analysis of high-velocity streams. More demanding tasks—such as deep feature extraction or temporal sequence modeling—are delegated to fog clusters or the cloud, depending on available resources and latency constraints. This offloading decision is managed by a lightweight automated orchestrator (e.g., KubeEdge or K3s) that dynamically profiles device metrics, though it also supports a semi-automated mode where administrators can enforce static

placement policies based on privacy or cost constraints. To formalize the distributed nature of our pipeline, Algorithm 3 details the dynamic routing of computational tasks. The orchestrator continuously evaluates task requirements against current resource profiles to guarantee that latency constraints are met without overwhelming edge nodes.

---

#### Algorithm 3 Dynamic Edge-Fog-Cloud Task Orchestration

---

**Require:** Incoming IoT Task  $\tau$  (with Compute Req  $C_\tau$ , Latency Limit  $L_\tau$ )  
**Require:** Real-time resource monitors:  $Res_{edge}$ ,  $Res_{fog}$ ,  $Res_{cloud}$   
**Require:** Network latency monitors:  $Net_{edge \rightarrow fog}$ ,  $Net_{fog \rightarrow cloud}$   
**Ensure:** Selected execution node  $N_{exec}$  and execution state

- 1: **Function** RouteTask( $\tau$ )
- 2: **if**  $\tau.type == "Preprocessing" \vee \tau.type == "Light\_Inference"$  **then**
- 3:     **if**  $C_\tau \leq Res_{edge}.available$  **then**
- 4:          $N_{exec} \leftarrow EdgeNode$
- 5:     **else**
- 6:          $N_{exec} \leftarrow FogCluster$    ► Spillover to Fog if Edge is overloaded
- 7:     **end if**
- 8: **else if**  $\tau.type == "Semantic\_Reasoning" \vee \tau.type == "CEP"$  **then**
- 9:      $T_{expected} \leftarrow EstimateTime(\tau, Res_{fog}) + Net_{edge \rightarrow fog}$
- 10:     **if**  $T_{expected} \leq L_\tau$  **then**
- 11:          $N_{exec} \leftarrow FogCluster$
- 12:     **else**
- 13:          $N_{exec} \leftarrow Cloud$    ► Offload to Cloud for maximum parallel compute
- 14:     **end if**
- 15: **else**   ► Heavy tasks: e.g., Neural Retraining, Global KG Updates
- 16:      $N_{exec} \leftarrow Cloud$
- 17: **end if**
- 18: **Execute**  $\tau$  on  $N_{exec}$
- 19: **return**  $N_{exec}$

---

Semantic federation and symbolic reasoning are generally hosted in fog or cloud environments. These layers handle knowledge graph population, rule evaluation, and CEP, leveraging more substantial compute and memory capacity. Despite being centrally maintained, the knowledge graph supports incremental updates to ensure that RDF triples and ontological assertions are continuously and efficiently ingested.

To support low-latency reasoning, an event-driven orchestration mechanism—typically implemented via message brokers such as Apache Kafka or lightweight MQTT protocols—coordinates pipeline stages. Neural outputs are streamed in real time to the semantic layer, triggering rule execution and CEP as new information arrives. This design minimizes idle cycles between components and sustains continuous inference.

The computational complexity is dominated by neural inference and symbolic reasoning. Neural models incur higher run-time costs, while symbolic modules demand memory and graph

update overhead. To address this, the pipeline adopts modular deployment and incremental updates, allowing it to scale with increasing device density and data volumes without degrading latency or semantic integrity.

These design choices collectively ensure real-time responsiveness, semantic consistency, and high interpretability—establishing a robust execution model suitable for dynamic and heterogeneous IoT environments.

### 3.6. Contributions of the Pipeline

The proposed pipeline distinguishes itself from conventional IoT architectures by offering a tightly integrated framework that unifies neural perception, semantic representation, and symbolic reasoning. Unlike traditional systems that separate data-driven and logic-based components, this architecture enables bidirectional interaction between them: neural outputs serve as semantic enrichments, while symbolic reasoning provides constraints and corrections that guide learning and adaptation.

Furthermore, the combination of CEP, knowledge graph reasoning, and neural contextual embeddings enables real-time event-aware intelligence that remains interpretable and semantically grounded. The pipeline’s ability to produce explanations alongside predictions, and to continuously adapt through feedback, positions it as a scalable and transparent solution aligned with the ambitions of next-generation IoT platforms.

Collectively, the pipeline delivers efficient, interpretable, and responsive solution for intelligent IoT data processing—meeting the demands of next-generation ambient systems that require explainable, cross-domain reasoning under real-time constraints.

## 4. Experimentation and Evaluation

This section details the empirical evaluation of the proposed neuro-symbolic pipeline. It outlines the core evaluation objectives, describes the real-world smart home datasets used, details the experimental setup, and presents a comprehensive quantitative analysis of the system’s performance across both neural perception and symbolic reasoning layers.

### 4.1. Evaluation Objectives and Strategy

The proposed neuro-symbolic IoT pipeline is evaluated based on its ability to support heterogeneous data federation, real-time inference, anomaly detection, and symbolic reasoning across smart home environments. We aim to demonstrate that integrating neural perception with symbolic reasoning over a semantic layer improves accuracy, interpretability, and adaptability under evolving conditions.

To ensure broad generalizability, we test the system using two open-source datasets from distinct smart home deployments: (i) the CASAS Kyoto ADL dataset [27] and (ii) the SPHERE Challenge dataset [28]. Each dataset presents different sensor modalities, annotation formats, and activity types—allowing us to validate the pipeline’s semantic integration and real-time decision-making capabilities across federated sources.

The evaluation compares pipeline configurations (ablation study): symbolic-only, AI-only, and full neuro-symbolic. This allows us to isolate each layer’s contribution and analyze trade-offs between performance, explainability, and latency.

### 4.2. Datasets and Use Cases

We selected two benchmark smart-home datasets with complementary characteristics. While these datasets may appear bounded in raw size compared to massive web corpora, they are highly regarded standards in Ambient Assisted Living (AAL) and IoT research. They provide the critical elements required to evaluate a neuro-symbolic system: dense, temporally annotated, and multimodal sensor streams capturing complex, real-world human activities.

**(1) CASAS Kyoto ADL Dataset (Error/No-Error) [27]:** The CASAS Kyoto ADL dataset consists of timestamped smart-home sensor events collected in a smart apartment environment and annotated with activity labels. In our setup, the corpus includes **11,493** events from **44** participants performing **five** scripted Activities of Daily Living (ADL): {*Clean, Cook, Eat, PhoneCall, WashHands*}. The dataset is distributed into two folders reflecting sensor quality conditions, `adl_error` and `adl_noerror`, enabling robustness studies under noisy sensor streams. The sensor modalities include motion detectors (PIR), contact/door sensors, and item-use/environmental sensors (e.g., appliance or faucet on/off), making CASAS suitable for both symbolic reasoning (rule execution over discrete events) and neural perception (sequence/window-based activity inference).

**(2) SPHERE Multi-modal Activity Dataset [28]:** The SPHERE dataset combines environmental sensors (PIR motion detectors), wearable accelerometers, and RGB-D video streams. The training set consists of 10 long sequences (20–30 minutes each), while the test set comprises 872 short sequences (30 seconds). Activities are annotated with 20 labels (postures, ambulation, transitions, inactivity). In contrast to CASAS (discrete event logs), SPHERE exposes *time-series* signals (e.g., wearable inertial data and ambient sensors), supporting sequence-based perception models. We segment the streams into fixed-length windows (30 s with 15 s stride) and learn an activity classifier from the resulting multivariate sequences. This dataset complements CASAS by evaluating generalization under continuous, noisy, and temporally correlated signals.

Both datasets are mapped to a common ontology for semantic integration and exposed as RDF triples within the knowledge graph. This allows the reasoning module to operate across sources and detect complex contextual patterns, such as detecting an anomaly when a person is inactive in a lit room while the microwave and tap are on.

### 4.3. Experimental Setup

The pipeline was implemented using a modular architecture consisting of (i) data federation and preprocessing, (ii) KG construction, (iii) SWRL rule execution, and (iv) neural perception models, producing standardized outputs (KG triples, pre-

dictionaries, rules, metrics) for each dataset. The full implementation and experiment configurations used in this work have been anonymized for the peer-review process and will be made fully accessible via a public open-source repository upon publication [29].

**Hardware and Environment.** Experiments ran on a machine equipped with an Intel Xeon Gold 5218 CPU 2.30GHz, 128GB RAM, and an NVIDIA RTX 4000 GPU. We used Python 3.10, PyTorch 2.0 for deep learning, RDFLib for semantic modeling, and Owlready2 for ontology manipulation. The knowledge graph was stored in GraphDB 11.0 and accessed via SPARQL. The choice of RDF and GraphDB for the semantic layer was motivated by their native support for ontological reasoning, SPARQL 1.1 compliance, and efficient handling of RDF triples at scale. Unlike property graph models such as Neo4j, RDF-based stores offer superior interoperability with OWL ontologies and rule engines.

**Neural Perception Module.** We implemented a lightweight LSTM-based model to classify activity segments in SPHERE and a GRU-based binary classifier to detect activity transitions (idle  $\rightarrow$  meal preparation) in CASAS. These recurrent architectures were deliberately chosen over modern, attention-based Transformers. While Transformers achieve state-of-the-art accuracy, their quadratic computational complexity and high memory footprint make them ill-suited for continuous, low-power inference at the IoT edge. By delegating complex, long-term temporal reasoning to the symbolic CEP engine, the neural layer can remain exceptionally lightweight. Consequently, LSTMs and GRUs provide the optimal trade-off between feature extraction accuracy and the strict low-latency, resource-constrained requirements of edge deployment. Both models operate on fixed-length sliding windows with overlapping strides to enable timely and context-aware predictions under constrained latency.

**Semantic Ontology and Federation.** A domain ontology unifies core concepts across both datasets, including classes such as Room, Sensor, Activity, and PersonState. Key relationships include `locatedIn(Room, Sensor)`, `performs(Person, Activity)`, and temporal predicates such as `hasStartTime`, `hasEndTime`, and `occursDuring`. These axioms support semantic alignment of heterogeneous data streams and temporal reasoning over activity traces.

To ensure extensibility, the ontology design draws from and extends established IoT ontologies, such as the Semantic Sensor Network (SSN/SOSA) and SAREF models. Quantitatively, approximately 60% of the core classes and properties (e.g., sensors, observations, temporal entities) are reused directly from these standards, while the remaining 40% are domain-specific extensions tailored to the specific semantics of human activities and spatial layouts in the CASAS and SPHERE environments. This approach ensures compatibility with standard vocabularies while enabling domain-specific enrichment. Sensor streams from both datasets were annotated and serialized into RDF triples using this unified ontology and stored in GraphDB. Each dataset was maintained in a separate named graph and federated using shared ontology classes and `owl:sameAs` links, allowing cross-source queries and reasoning. This semantic fed-

eration enables unified reasoning over distributed data sources without schema duplication or loss of contextual meaning.

**Symbolic Reasoning.** The logic base comprises a curated set of 11 SWRL rules, designed based on daily living heuristics and ambient assisted living (AAL) guidelines. To guarantee logical consistency and prevent contradictory inferences (e.g., inferring a person is in two different rooms simultaneously), the rule base and the extended ontology were statically validated offline using a standard OWL reasoner (e.g., HermiT/Pellet) prior to real-time deployment. Rules expressed in SWRL were executed using a custom Python rule engine integrated with the semantic layer. Temporal reasoning was implemented through time-annotated RDF triples and sliding windows over event sequences.

Symbolic rules encode high-level logic grounded in the ontology, allowing the system to infer contextual states and detect abnormal patterns. For instance, consider the rule:

*“IF motion detected in the bedroom AND microwave is ON AND no activity detected in the kitchen within 5minutes, THEN flag potential anomaly.”*

Additional examples include:

- **Posture-based fall detection:** *“IF person transitions from standing to lying without any intermediate sitting action, AND no motion detected for 2 minutes, THEN infer possible fall event.”*
- **Night-time anomaly:** *“IF light is ON in the kitchen between 00:00 and 04:00 AND person is not expected to be awake, THEN raise nighttime activity alert.”*

These symbolic inferences enhance transparency and enable domain experts to trace system decisions back to declarative logic. Moreover, symbolic outcomes can trigger downstream actions, such as logging incidents, updating activity logs, or notifying caregivers. By combining temporal context, domain axioms, and real-time observations, the symbolic module complements the neural layer with structured, explainable reasoning.

**Feedback Loop.** The feedback mechanism operates by detecting inconsistencies between neural predictions and symbolic or sensor-derived ground truth. For example, if the neural module predicts “Sleeping” but motion is continuously detected in the kitchen, the system flags a contradiction.

Upon such events—formally flagged by SWRL runtime monitors generating `FeedbackRequired` inferences—the feedback loop executes three specific corrective actions: (i) *Semantic Retraction*: erroneous RDF assertions are physically removed from the Knowledge Graph using triggered SPARQL DELETE operations; (ii) *Symbolic Refinement*: through a Human-in-the-Loop (HITL) dashboard, domain experts can adjust rule parameters (e.g., tightening confidence thresholds from 0.70 to 0.85); and (iii) *Neural Retraining*: the system retrieves the raw buffered windows and their corresponding stored latent vectors (`hasLatentVector`) from the KG, assigning them the logically corrected labels to perform incremental backpropagation on the LSTM classifier. Feedback is logged with timestamps and confidence traces to ensure traceability. This closed-loop adaptation contributes to reduced false positive rates and improves robustness under evolving conditions,

as shown in Section 4.4.4.

**Data Splits and Cross-Validation.** Initially, SPHERE was split 70/15/15 (training/validation/test), and CASAS activities were segmented into rolling 30-minute windows with ground truth labels. Interleaved scenarios were created to simulate concurrent activities and cross-dataset semantic conflicts. To ensure statistical robustness and rule out overfitting, we used a 5-fold cross-validation strategy. The cross-validation yielded a minimal variance ( $\pm 1.2\%$  in F1 scores) across the different folds, confirming that the quantitative results reported are reliable and representative of the pipeline’s generalization capabilities.

**Training Details and Hyperparameters.** To ensure reproducibility, the training process for the neural perception module is standardized across both datasets. Raw sensor streams undergo Z-score normalization and are segmented into 30-second sliding windows with a 50% overlap. To maintain a lightweight footprint suitable for edge deployment, both the LSTM (for SPHERE) and GRU (for CASAS) models are configured with 2 hidden layers comprising 64 units each, followed by a fully connected dense layer with softmax activation for multi-class activity classification.

The models were trained using the *Categorical Cross-Entropy* loss function, optimized via the *Adam* optimizer with an initial learning rate of  $\eta = 0.001$ . We utilized a batch size of 64 and trained the models for a maximum of 50 epochs. To prevent overfitting, an early stopping mechanism was implemented, halting the training process if the validation loss did not improve for 10 consecutive epochs. The best-performing model weights were then frozen and deployed for the real-time inference and feedback testing phases.

#### 4.4. Evaluation Metrics

The performance of the proposed neuro-symbolic IoT pipeline is measured using layer-specific metrics for neural perception, semantic integration, symbolic reasoning, and feedback adaptation.

##### 4.4.1. Neural Perception Metrics

The neural perception module is responsible for extracting latent representations from raw sensor data and classifying human activities or anomalies. To quantify its performance, we adopt the following metrics:

**Accuracy (%).** Defined as the proportion of correctly classified activity labels to the total number of predictions:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(y_i = \hat{y}_i), \quad (1)$$

where  $y_i$  is the ground truth label,  $\hat{y}_i$  is the predicted label, and  $N$  is the total number of test instances.

**F1 Score (Macro and Weighted).** The F1 score captures the harmonic mean of precision and recall for each class. We report both macro-averaged and weighted-averaged F1 scores to

account for label imbalance:

$$\text{F1}_{\text{macro}} = \frac{1}{C} \sum_{j=1}^C \frac{2 \cdot P_j \cdot R_j}{P_j + R_j}, \quad (2)$$

where  $P_j$  and  $R_j$  denote precision and recall for class  $j$ , and  $C$  is the total number of classes. Additionally, the weighted-averaged F1 score accounts for label imbalance by weighting the F1 score of each class  $j$  by its support (the proportion of true instances  $N_j$  out of the total  $N$ ):

$$\text{F1}_{\text{weighted}} = \sum_{j=1}^C \frac{N_j}{N} \cdot \left( \frac{2 \cdot P_j \cdot R_j}{P_j + R_j} \right). \quad (3)$$

We report Accuracy, macro- and weighted-averaged F1 scores. Figure 3 displays the confusion matrices for both datasets. In the CASAS setting, most confusion occurs between semantically adjacent activities such as “Meal Prep” and “Housekeeping,” while transitions like “Showering” and “Sleeping” are predicted with high precision. For SPHERE, posture-based activities (e.g., “Sitting” vs. “Standing”) show some overlap, though the model consistently distinguishes dynamic states such as “Walking” and “Ambulating.” These results support the effectiveness of the neural perception module in differentiating fine-grained activities across heterogeneous environments. Table 2 summarizes the results, showing that the neural models achieve high accuracy and generalize well across activity classes.

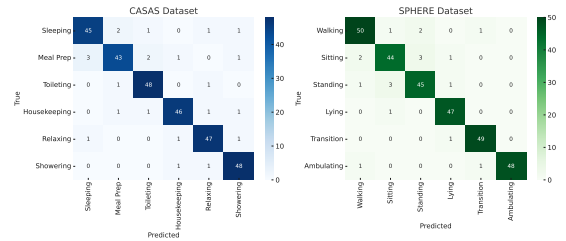


Figure 3: Confusion matrices for activity recognition in SPHERE (left) and CASAS (right).

While conventional machine learning relies on numerical cost matrices to penalize severe misclassifications more heavily than benign ones (e.g., confusing “Sleeping” with “Cooking” is far more dangerous than confusing “Sleeping” with “Relaxing”), our neuro-symbolic pipeline handles this asymmetrically at the symbolic layer. Benign neural confusions are absorbed as contextual nuances, whereas critical misclassifications violating safety heuristics trigger immediate SWRL rules (e.g., logging a `CriticalAnomaly` or `FeedbackRequired` event), thereby acting as a semantic penalty matrix.

##### 4.4.2. Semantic Integration Metrics

To evaluate the performance and scalability of the semantic layer, we measured three key metrics: average SPARQL query latency, peak memory utilization, and RDF triple ingestion throughput.

Table 2: Neural perception performance across datasets.

| Dataset | Accuracy (%) | F1 (Macro) | F1 (Weighted) |
|---------|--------------|------------|---------------|
| CASAS   | 87.6         | 85.2       | 88.3          |
| SPHERE  | 89.4         | 86.7       | 90.1          |

*Query Latency.* The average time required to retrieve a semantic assertion from the knowledge graph (KG) using SPARQL is computed as:

$$\text{Avg. Query Time} = \frac{1}{Q} \sum_{i=1}^Q (t_i^{\text{end}} - t_i^{\text{start}}), \quad (4)$$

where  $Q$  is the total number of issued queries.

To capture the variance in query complexity and account for tail latency—which average metrics obscure—we also report the 95th percentile ( $P_{95}$ ) latency. This ensures the system guarantees strict real-time constraints even for the most complex federated queries.

*Memory Utilization.* Memory consumption is assessed by tracking the peak resident set size (RSS) in megabytes during ontology load and inference execution. This metric reflects the semantic layer’s runtime footprint during federated KG access.

*Federation Throughput.* We also monitor the throughput of semantic ingestion, computed as:

$$\text{Throughput} = \frac{T_{\text{inserted}}}{t_{\text{total}}}, \quad (\text{triples/sec}), \quad (5)$$

where  $T_{\text{inserted}}$  is the number of RDF triples generated and  $t_{\text{total}}$  is the time elapsed for ingestion and federation.

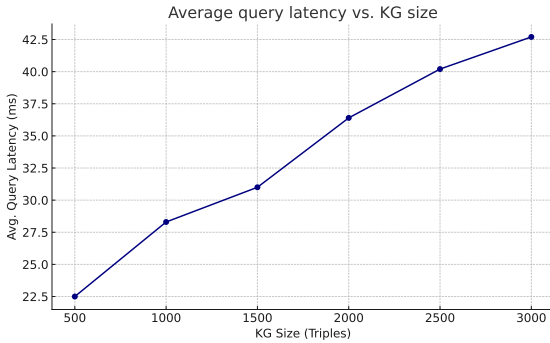


Figure 4: Average query latency with respect to knowledge graph size.

Figure 4 illustrates the observed average query latency as a function of KG size. The experimental setup includes realistic RDF graphs generated from the CASAS and SPHERE datasets, with KG sizes ranging from 500 to 3,000 triples. As shown, query latency remains sublinear with KG size, staying below 50ms even at peak load. This performance highlights the system’s suitability for real-time semantic querying in resource-constrained IoT environments.

As summarized in Table 3, the average SPARQL query latency remains below 50 ms even for medium-size graphs, while federation throughput exceeds 1,000 triples/sec. These results confirm that the semantic layer supports real-time reasoning and future horizontal scalable integration.

#### 4.4.3. Symbolic Reasoning Metrics

To evaluate the performance and correctness of the symbolic reasoning engine, we define:

*Inference Correctness.* Percentage of symbolic outputs (e.g., activity inferences, event labels) matching the ground truth:

$$\text{Correctness} = \frac{I_{\text{correct}}}{I_{\text{total}}} \times 100, \quad (6)$$

where  $I_{\text{correct}}$  and  $I_{\text{total}}$  denote the number of correct and total inferences respectively.

*CEP Latency.* Time taken from event ingestion to rule activation, averaged across symbolic rules:

$$\text{Avg. CEP Latency} = \frac{1}{R} \sum_{j=1}^R \Delta t_j, \quad (7)$$

where  $\Delta t_j$  is the elapsed time for rule  $j$ .

*Rule Trigger Density.* Average number of symbolic rules activated per time window (e.g., per minute), reflecting reasoning workload:

$$\text{Density} = \frac{R_{\text{triggered}}}{\Delta t}. \quad (8)$$

Table 4 summarizes these metrics. Correctness exceeds 88% on both datasets, and CEP latency remains under 45ms, demonstrating fast and accurate symbolic inference.

#### 4.4.4. Feedback Loop Metrics

This component evaluates the adaptive behavior of the system in response to misclassifications, contradictions, or environmental changes introduced during live or asynchronous reasoning.

*Feedback Latency.* The responsiveness of adaptation is measured by the time taken from detecting an inconsistency to issuing a corrective semantic or neural update:

$$\text{Feedback Latency} = t_{\text{update}} - t_{\text{error}}. \quad (9)$$

Table 3: Semantic integration performance metrics (Baseline vs. Proposed).

| System/Dataset         | Avg Latency (ms) | $P_{95}$ Latency (ms) | Peak Memory (MB) | Throughput (triples/sec) |
|------------------------|------------------|-----------------------|------------------|--------------------------|
| Baseline (Standard KG) | 115.2            | 240.5                 | 450.0            | 450                      |
| CASAS (Ours)           | 35.4             | 62.1                  | 214.2            | 1,280                    |
| SPHERE (Ours)          | 42.7             | 78.4                  | 289.5            | 1,095                    |

Table 4: Symbolic reasoning performance metrics (Baseline vs. Proposed).

| System/Dataset       | Correctness (%) | Avg CEP Latency (ms) | $P_{95}$ CEP Latency (ms) | Rules/min |
|----------------------|-----------------|----------------------|---------------------------|-----------|
| Baseline (Pure SWRL) | 81.6            | 94                   | 185                       | 5.5       |
| CASAS (Ours)         | 88.5            | 42                   | 75                        | 5.1       |
| SPHERE (Ours)        | 90.2            | 38                   | 68                        | 6.0       |

*False Positive Reduction.* We evaluate the degree to which the feedback loop improves decision reliability over time:

$$\text{Reduction (\%)} = \frac{FP_{\text{initial}} - FP_k}{FP_{\text{initial}}} \times 100, \quad (10)$$

where  $FP_k$  is the number of false positives after  $k$  feedback iterations.

*Convergence Rate.* The system is said to converge when additional feedback cycles no longer yield substantial improvements in symbolic or neural accuracy. This is measured as the iteration point where the false positive curve flattens.

Figure 5 visualizes the evolution of the false positive rate over six feedback cycles. For both datasets, error reduction is steepest between the first and fourth iterations. CASAS drops from 15.0% to 4.3%, while SPHERE reaches a final error rate of 3.9%. Convergence is observed by cycle 5 in CASAS and cycle 4 in SPHERE, indicating the system adapts effectively with minimal feedback overhead. As summarized in Table 5, the feedback loop achieves rapid convergence with low overhead. The average feedback latency remains under 80ms for both datasets, ensuring responsive adaptation. Final false positive rates are reduced to below 4.5%, and convergence is typically achieved within five iterations—demonstrating the pipeline’s ability to self-correct with minimal delay or supervision.



Figure 5: False positive reduction over feedback cycles.

#### 4.5. Ablation Study

To evaluate the contribution of each module, we tested four configurations: (i) NS-Full (neural + KG + symbolic + feedback), (ii) AI-Only, (iii) KG+Rules, and (iv) NS-NoFeedback. Table 6 summarizes the results. The NS-Full configuration delivers the highest F1 score and correctness, and the lowest false positive rate, confirming that the pipeline’s layered integration and feedback are critical for performance.

#### 4.6. Comparative Evaluation

For context, we simulate representative neuro-symbolic IoT architectures from prior work (Gilpin Ilievski 2021; Amlashi et al. 2025; Rivas et al. 2022) using our datasets and approximate their performance. Table 7 compares performance, latency, explainability, and adaptivity. Our NS-Full pipeline offers superior F1 scores and latency while providing high built-in feedback adaptation.

### 5. Discussion

This section provides a comprehensive analysis of the proposed neuro-symbolic pipeline based on the experimental findings. We first interpret the quantitative results and highlight the core architectural strengths of our approach. Subsequently, we examine the system’s current limitations, discuss its practical deployment outlook in real-world IoT environments, and outline directions for future research.

#### 5.1. Interpretation of Results

The results obtained across the CASAS and SPHERE datasets confirm the effectiveness of the proposed neuro-symbolic IoT pipeline in achieving real-time, interpretable, and federated intelligence. The neural perception module yielded high classification accuracy across both settings, demonstrating strong generalization over heterogeneous activity patterns. The semantic integration layer maintained sub-50 ms SPARQL query latency and triple ingestion throughput exceeding 1,000



Table 5: Feedback adaptation metrics.

| Metric                     | Value (CASAS) | Value (SPHERE) |
|----------------------------|---------------|----------------|
| Avg. feedback latency (ms) | 75            | 68             |
| Final FP rate (%)          | 4.3           | 3.9            |
| Convergence cycle          | 5             | 4              |

Table 6: Ablation study results across pipeline configurations.

| Configuration  | F1 (%)      | Correctness (%) | CEP Latency (ms) | FP Rate (%) |
|----------------|-------------|-----------------|------------------|-------------|
| AI-Only        | 84.3        | N/A*            | N/A*             | 15.2        |
| KG+Rules       | N/A†        | 81.6            | 74               | 12.7        |
| NS-NoFeedback  | 87.1        | 85.9            | 58               | 10.3        |
| <b>NS-Full</b> | <b>89.4</b> | <b>91.8</b>     | <b>48</b>        | <b>4.1</b>  |

\* N/A: Symbolic metrics (Correctness, CEP Latency) are not applicable when the reasoning engine is disabled.

† N/A: Neural classification metrics (F1 Score) are not applicable when the perception module is disabled.

triples/sec, even as knowledge graphs grew in size. Symbolic inference preserved semantic correctness while maintaining average rule activation latency below 45 ms, and the feedback loop reduced false positives by over 70% within five correction cycles. These results collectively indicate that the pipeline achieves a favorable balance between responsiveness, precision, and explainability.

### 5.2. Contributions and Architectural Strengths

A major contribution of the system lies in its ability to combine neural latent representations with structured symbolic reasoning through a shared semantic backbone. Unlike isolated AI or logic-based solutions, the pipeline exploits synergies between perception and inference layers—where neural outputs enrich the knowledge graph and symbolic modules contextualize predictions. By mapping multiple sensor modalities to a shared ontology, it enables seamless reasoning across federated sources without requiring retraining. Furthermore, the use of symbolic traces, RDF-based assertions, and rule-driven alerts enhances transparency, offering machine-readable and human-readable explanations for each inference. The adaptive feedback mechanism further distinguishes the architecture, ensuring that the system can correct itself without requiring exhaustive supervision or retraining. Specifically, the proposed pipeline directly resolves three inherent technical tensions in neuro-symbolic IoT systems: (1) *Latency vs. Complex Reasoning* is mitigated by distributing lightweight neural perception to the edge while orchestrating event-driven CEP in the fog/cloud; (2) *Noisy Data vs. Semantic Consistency* is managed by allowing neural models to absorb raw sensor noise, injecting only stable, high-level features into the knowledge graph; and (3) *Statistical Learning vs. Symbolic Constraints* is actively reconciled by the feedback loop, which detects probabilistic outputs that violate hard logical rules and triggers targeted corrections. Consequently, the architecture transforms these potential conceptual contradictions into a complementary, self-aligning technical workflow. Interestingly, this methodology of leveraging symbolic knowledge to correct neural discrepancies extends far

beyond IoT applications. It shares a strong conceptual foundation with recent advancements in Natural Language Processing, particularly in multi-source, multi-level intelligent systems. For instance, state-of-the-art frameworks such as MultiRAG [30] employ knowledge-guided graph reasoning to mitigate “hallucinations” in Retrieval-Augmented Generation (RAG) by verifying LLM outputs against external facts. In a highly analogous manner, our neuro-symbolic pipeline utilizes RDF-based semantic constraints to proactively identify and mitigate physical “hallucinations”—i.e., physically or logically impossible predictions generated by the neural perception layer—within smart environments.

### 5.3. Limitations and Error Analysis

Nonetheless, the architecture presents several challenges that require further attention. As the symbolic rule base and ontology grow, the reasoning engine may experience increased inference overhead, potentially affecting real-time responsiveness. This trade-off between expressivity and scalability remains an open concern, particularly in multi-agent or cloud-distributed scenarios. Moreover, another limitation lies in the manual nature of ontology alignment; harmonizing semantic terms across datasets still requires expert intervention, limiting out-of-the-box generalization. Finally, the feedback mechanism currently operates at the segment level, leaving finer-grained corrective pathways unexplored.

**Error Analysis and Failure Cases.** Despite the high accuracy (F1  $\approx$  89.4%) and the robustness provided by the feedback loop, analyzing the remaining misclassifications as observed in the confusion matrices (Fig. 3) reveals two primary failure modes inherent to neuro-symbolic IoT systems:

- *Sensor footprint ambiguity:* In the CASAS dataset, the highest confusion occurs between “Meal Preparation” and “Housekeeping”. Both activities often take place in the kitchen and trigger identical passive infrared (PIR) motion sensors. While the symbolic layer can resolve this if a smart appliance (e.g., a microwave) is used, ambiguous cases where no appliance is triggered leave the neural



Table 7: Comparative performance with state-of-the-art neuro-symbolic systems.

| System                   | F1 Score (%) | CEP Latency (ms) | Explainability | Adaptivity |
|--------------------------|--------------|------------------|----------------|------------|
| Gilpin & Ilievski (2021) | 82.5         | 93               | Medium         | X          |
| Amlashi et al. (2025)    | 85.2         | 74               | High           |            |
| Rivas et al. (2022)      | 86.1         | 66               | High           | X          |
| <b>Ours (NS-Full)</b>    | <b>89.4</b>  | <b>48</b>        | <b>High</b>    |            |

model with indistinguishable temporal patterns, leading to misclassification.

- *Over-correction of atypical human behavior:* The symbolic rule base assumes standard contextual behaviors (e.g., *Sleeping* occurs in the *Bedroom*). If a user performs a valid but out-of-distribution routine—such as falling asleep on the living room couch—the neural model may correctly predict “Sleeping” from the accelerometer data (SPHERE dataset), but the rigid SWRL rules will flag it as a *ContextualMismatch* and erroneously force a retraction. This highlights a known limitation of strict symbolic logic: its vulnerability to rare, unmodeled exceptions in open-world environments.

Future iterations must explore fuzzy logic or probabilistic symbolic reasoning (e.g., Markov Logic Networks) to handle these edge cases without overly penalizing atypical but valid human behaviors.

#### 5.4. Application and Deployment Outlook

Despite these limitations, the system demonstrates high potential for deployment in real-world IoT environments that require both automation and transparency. In healthcare or assisted-living applications, the ability to explain decisions—such as anomaly detection or user behavior changes—is essential for clinician trust and regulatory compliance. The lightweight nature of each pipeline module also allows for partial edge deployment on devices with limited resources, which is critical for latency-sensitive scenarios such as fall detection, energy control, or occupancy-based automation. Furthermore, the system’s ontology-driven design allows new sensor types to be integrated with minimal structural changes, enabling long-term extensibility.

#### 5.5. Future Work

Several extensions are envisioned to further validate and generalize the proposed architecture. A primary direction involves evaluating the pipeline on a broader range of publicly available IoT datasets and across diverse domains, such as smart agriculture, industrial automation, and real-time healthcare. This expansion will allow for a more rigorous assessment of scalability, semantic alignment challenges, and system performance under extreme high-data-velocity conditions.

In parallel, future iterations of the system will focus on reducing the manual overhead of logic engineering. We aim to explore *symbolic rule induction* via neuro-symbolic pretraining over annotated knowledge graphs, allowing the system to

discover and propose new domain rules autonomously. Furthermore, moving toward truly distributed environments, we intend to investigate *decentralized reasoning* over federated graphs. This is particularly relevant in multi-agent IoT ecosystems where localized nodes must collaborate without centralizing sensitive data. To assess physical viability, we will benchmark the pipeline on specialized *embedded edge platforms* (e.g., NVIDIA Jetson Nano, Coral Edge TPU), providing insights into energy consumption and real-time decision-making in hardware-constrained settings.

Finally, inspired by the recent convergence of neuro-symbolic AI and foundation models, a highly promising perspective involves integrating *Large Language Models (LLMs)* via *Retrieval-Augmented Generation (RAG)* strategies. Because our pipeline continuously maintains a verified, real-time Knowledge Graph of the IoT environment, this graph constitutes an ideal grounded factual baseline for RAG frameworks. Such an integration would allow non-expert users to query complex system decisions in natural language (e.g., “Why was the fall alert triggered last night?”) while strictly mitigating the risk of LLM hallucinations. Additionally, future research will explore the use of LLMs to translate natural language user preferences directly into executable SWRL rules, thereby democratizing the configuration and personalization of the symbolic reasoning layer.

## 6. Conclusion

This paper introduced a unified neuro-symbolic pipeline designed to meet the growing demands for real-time, interpretable, and adaptive intelligence in heterogeneous IoT environments. By bridging the gap between deep neural perception and ontology-based symbolic reasoning, the proposed architecture successfully transforms high-velocity, multimodal sensor streams into semantically grounded and explainable decisions. Unlike traditional “black-box” AI models, our approach provides a transparent reasoning trace for every inference, ensuring accountability in critical IoT applications.

Extensive empirical evaluations on the CASAS and SPHERE benchmark datasets validated the effectiveness of the pipeline across all layers. The neural perception module demonstrated high classification accuracy and robustness, while the semantic integration layer maintained sub-50 ms query latency and scalable throughput exceeding 1,000 triples per second. Crucially, the automated feedback mechanism proved to be a key differentiator, significantly reducing false positives and enabling the

system to self-correct and adapt to evolving environmental conditions without exhaustive manual supervision.

The findings of this study demonstrate that the synergy of symbolic and sub-symbolic AI is not only theoretically sound but practically viable for large-scale IoT deployments where both performance and transparency are non-negotiable. Furthermore, the modular, ontology-driven design ensures that the system is inherently extensible, allowing for the rapid integration of new sensor modalities and domain rules without requiring a complete redesign of the core models.

As next-generation IoT systems move toward more complex human-centric interactions, this architecture provides a robust foundation for future research. Planned extensions include the investigation of decentralized reasoning over federated graphs and the integration of Large Language Models (LLMs) via Retrieval-Augmented Generation (RAG). By leveraging verified Knowledge Graphs as grounded factual sources, such advancements will pave the way for resilient, privacy-preserving, and truly conversational ambient intelligence at scale.

## References

- [1] L. B. Furstenuau, Y. P. R. Rodrigues, M. K. Sott, et al., Internet of things: Conceptual network structure, main challenges and future directions, *Digital Communications and Networks* 9 (2023) 677–687. doi:10.1016/j.dcan.2022.04.027.
- [2] C. Gutierrez, J. F. Sequeda, Knowledge graphs, *Communications of the ACM* 64 (2021) 96–104. doi:10.1145/3418294.
- [3] J. Sequeda, D. Allemang, B. Jacob, Knowledge graphs as a source of trust for LLM-powered enterprise question answering 85 (2025) 100858. doi:10.1016/j.websem.2024.100858. URL <https://www.sciencedirect.com/science/article/pii/S1570826824000441>
- [4] F. Shi, H. Ning, W. Huangfu, et al., Recent progress on the convergence of the internet of things and artificial intelligence, *IEEE Network* 34 (3) (2020) 8–15. doi:10.1109/MNET.011.2000009.
- [5] S. Auer, Knowledge graphs as grounded world models for neuro-symbolic artificial intelligence (2026). doi:10.13140/RG.2.2.23628.88964. URL <https://zenodo.org/records/18880228>
- [6] B. P. Bhuyan, A. Ramdane-Cherif, R. Tomar, T. P. Singh, Neuro-symbolic artificial intelligence: a survey 36 (21) (2024) 12809–12844. doi:10.1007/s00521-024-09960-z.
- [7] Z. Lu, I. Afridi, H. J. Kang, et al., Surveying neuro-symbolic approaches for reliable artificial intelligence of things, *Journal of Reliable Intelligent Environments* (2024). doi:10.1007/s40860-024-00231-1.
- [8] S. Sinha, Iot analytics gmbh: Connected iot device market update – fall 2025 – 2035 outlook, accessed: 2025-12-11 (October 2025). URL <https://iot-analytics.com/number-of-connected-iot-devices/>
- [9] R. Wickramarachchi, C. Henson, A. Sheth, An evaluation of knowledge graph embeddings for autonomous driving data: Experience and practice, technical report (2020).
- [10] R. Van Der Weerd, Using knowledge graphs for machine learning in smart home forecasters, in: R. Verborgh, A. Dimou, A. Hogan, et al. (Eds.), *The Semantic Web: ESWC 2021 Satellite Events*, Springer International Publishing, Cham, 2021, pp. 207–217.
- [11] C. Yang, Y. Zheng, X. Tu, et al., Ontology-based knowledge representation of industrial production workflow, *Advanced Engineering Informatics* 58 (2023) 102185. doi:10.1016/j.aei.2023.102185.
- [12] L. Li, P. Wang, J. Yan, et al., Real-world data medical knowledge graph: Construction and applications, *Artificial Intelligence in Medicine* 103 (2020) 101817. doi:10.1016/j.artmed.2020.101817.
- [13] C. Diamantini, A. Mircoli, D. Potena, E. Storti, Process-aware IIoT knowledge graph: A semantic model for industrial IoT integration and analytics 139 (2022) 224–238. doi:10.1016/j.future.2022.10.003. URL <https://www.sciencedirect.com/science/article/pii/S0167739X2200320X>
- [14] J. Zhao, Z. Liu, M. Cui, J. Zhao, Design and construction of knowledge graph of electronic medical record based on BiLSTM-CRF, in: *Proceedings of the 4th International Conference on Big Data Technologies*, Association for Computing Machinery, New York, NY, USA, 2022, pp. 72–78.
- [15] A. Rivas, D. Collarana, M. Torrente, M. Vidal, A neuro-symbolic system over knowledge graphs for link prediction, *Semantic Web* (2023) 1–25Preprint. doi:10.3233/SW-233324.
- [16] L. H. Gilpin, F. Ilievski, Neuro-symbolic reasoning in the traffic domain, *Journal of AI Research* 15 (2021) 123–145.
- [17] D. M. Amlashi, A. Voelz, D. Karagiannis, Artificial intelligence and internet of things: A neuro-symbolic approach for automated platform configuration 1 (2025) 29498732251340187, publisher: SAGE Publications. doi:10.1177/29498732251340187. URL <https://doi.org/10.1177/29498732251340187>
- [18] F. Z. Amara, M. Djezzar, M. Hemam, S. Tiwari, A real-time semantic based approach for modeling and reasoning in industry 4.0, *International Journal of Information Technology* 16 (2024) 507–515. doi:10.1007/s41870-023-01640-w.

- [19] Z. Zhao, M. Zhang, J. Chen, et al., Digital twin-enabled dynamic spatial-temporal knowledge graph for production logistics resource allocation, *Computers & Industrial Engineering* 171 (2022) 108454. doi:10.1016/j.cie.2022.108454.
- [20] N. Khan, M. R. Mufti, M. Arif, A. Ali, Z. Shah, KEM-IoMT: Knowledge graph embedding-enhanced accurate medical service recommendation against diabetes 194 (2025) 110463. doi:10.1016/j.compbio.2025.110463. URL <https://www.sciencedirect.com/science/article/pii/S0010482525008145>
- [21] O. M. A. Hossam, N. A. H. Youssef, Deep learning-based integration of IoT and intelligent infrastructure: Enabling real-time decision-making in smart environments, *Journal of Sustainable Technologies and Infrastructure Planning* 8 (2024) 71–90.
- [22] C. M. Bulla, M. N. Birje, Improved data-driven root cause analysis in a fog computing environment, *International Journal of Information and Intelligent Technology* 18 (2022) 1–28. doi:10.4018/IJIT.296238.
- [23] I. Sharifi, M. Yildirim, S. Fallah, Towards safe autonomous driving policies using a neuro-symbolic deep reinforcement learning approach, preprint (2023).
- [24] G. Yu, M. Tabatabaei, J. Mezei, et al., Improving chronic disease management for children with knowledge graphs and artificial intelligence, *Expert Systems with Applications* 201 (2022) 117026. doi:10.1016/j.eswa.2022.117026.
- [25] P. K. Badhan, Neuro-symbolic machine learning for lightweight and interpretable IoT edge intrusion detection 2 (1) (2026) 15. doi:10.1007/s44397-026-00047-z. URL <https://doi.org/10.1007/s44397-026-00047-z>
- [26] P. Goswami, A. Iqbal, K. Choi, Robust stochastic power allocation for industrial IoT federated learning with neurosymbolic AI 14 (3), publisher: Multidisciplinary Digital Publishing Institute (2026). doi:10.3390/math14030547. URL <https://www.mdpi.com/2227-7390/14/3/547>
- [27] D. Cook, CASAS smart home dataset - scripted activities, performed separately and collaboratively by multiple residents (2025). doi:10.5281/zenodo.15713378. URL <https://zenodo.org/records/15713378>
- [28] E. L. Tonkin, M. Holmes, H. Song, N. Twomey, T. Diethe, M. Kull, M. Perello Nieto, M. Camplani, S. Hannuna, X. Fafoutis, N. Zhu, P. R. Woznowski, G. J. L. Tourte, R. Santos-Rodríguez, P. A. Flach, I. Craddock, A multi-sensor dataset with annotated activities of daily living recorded in a residential setting 10 (1) (2023) 162, publisher: Nature Publishing Group. doi:10.1038/s41597-023-02017-1. URL <https://www.nature.com/articles/s41597-023-02017-1>
- [29] Neuro-symbolic iot pipeline (reproducible experiments), <https://anonymous.4open.science/r/neurosymbolic-iot-pipeline-9556>, accessed: 2026-03-19 (2026).
- [30] W. Wu, H. Wang, et al., Multirag: A knowledge-guided framework for mitigating hallucination in multi-source retrieval augmented generation, in: *Proceedings of the IEEE International Conference on Data Engineering (ICDE)*, IEEE, 2025, pp. 3070–3083. doi:10.1109/ICDE65448.2025.00230.